

Hiring

Interview Question:

Custom Employee Portal with Zoho One Integration

Note: Create Free trial account anywhere required

Project Objective

Build a custom employee portal with its own authentication and role-based access control (RBAC) system. The portal should integrate with Zoho One APIs and provide employees access only to the Zoho services they are authorized to use.

Business Requirements

1. Custom Employee Portal

Develop a web-based frontend application that serves as a centralized portal for all employees.

Features:

- Custom login and authentication system.
- Employee dashboard after login.
- Modern, responsive UI.
- Single entry point to all authorized Zoho services.

2. Role-Based Access Control (RBAC)

Implement a comprehensive RBAC system.

Roles should include:

Admin

- Full access to the portal.
- Access to all integrated Zoho One services.
- Create, edit, and delete users.
- Create and manage roles.
- Assign permissions to users and roles.
- Monitor user activity and access logs.

Manager (Optional)

- Access to assigned departments/functions.
- View team-related information and reports.

Employee

- Access only to Zoho applications and features assigned to their role.

3. Zoho One API Integration

The portal should integrate with Zoho One and connected Zoho applications through APIs.

Requirements:

- Use a single Zoho One account (or service account) for backend integrations.
- Employees should not require individual Zoho usernames or passwords.
- Access to Zoho services should be controlled entirely through the custom RBAC system.
- Only authorized services should be displayed in the employee dashboard.
- Backend should securely handle Zoho authentication tokens and API communication.

Examples:

- The employee portal must support role-based access and redirection to the following Zoho applications, based on the logged-in user's assigned role:

User Role	Zoho Application	Purpose
HR	Zoho People	HR management functions

Sales	Zoho CRM	Sales and customer relationship management
Support	Zoho Desk	Support ticketing and case management
Finance	Zoho Books	Financial and accounting operations

Expected behavior:

Each user should only see and be able to access the Zoho application relevant to their role. For example, if an employee logs in as an **Accountant (Finance role)**, they should see access to **Zoho Books**, and clicking on it should navigate them to the Zoho Books portal. Similarly, an **HR** user should see access to **Zoho People**, and clicking on it should navigate them to the Zoho People portal and so on for Sales (Zoho CRM) and Support (Zoho Desk).

- JWT-based authentication.
- Role and permission validation on every API request.
- Secure storage of Zoho OAuth tokens.
- Audit logs for login and activity tracking.
- Session management and timeout controls.
- HTTPS enforced across all environments.

5. Technical Architecture

Frontend

- React.js / Next.js
- Responsive UI
- Dashboard-based navigation

Backend

- Node.js (Express/NestJS)
- RBAC authorization engine
- Zoho API integration layer
- Authentication service

Database

- PostgreSQL / MySQL - For demo use any database language

Tables:

- Users
- Roles
- Permissions
- UserRoles
- RolePermissions
- AuditLogs

6. User Flow

1. Employee visits the custom portal.
2. Employee logs in using portal credentials.
3. System authenticates user.
4. System determines assigned roles and permissions.
5. Dashboard displays only authorized Zoho services.
6. Employee accesses Zoho data/functions through backend APIs without seeing any Zoho credentials.
7. Admin manages users, roles, and permissions from an admin dashboard.